

Sunanda Tata

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Education

California State University, Fullerton (Fullerton, CA) 08/2024 – 05/2026
Master of Science in Computer Science **GPA: 3.7 / 4.0**

Experience

Pacific Life (Newport Beach, CA) 06/2025 – 03/2026
Software Developer (Contract)

- Integrated **LangChain**-based validation workflows into loan ingestion pipelines to identify anomalies and rule violations, surfacing data-quality issues before downstream processing and reducing unnecessary operational escalations.
- Optimized API p95 latency by **40%** on **Kubernetes**-hosted services through **Redis** caching for frequently accessed loan data and **Kafka** consumer tuning for higher-throughput processing.
- Engineered loan portfolio validation services using **Java** and **Spring Boot**, exposing **REST APIs** consumed by **React** dashboards for ingestion status, compliance failures, and loan-level reporting across **200+ internal users**.
- Delivered **TypeScript** components for loan approvals and exception-handling workflows through bi-weekly **CI/CD** releases with **GitHub Actions**, participating in peer code reviews and adhering to **WCAG 2.1 AA** accessibility standards.

California State University, Fullerton (Fullerton, CA) 09/2024 – 05/2025
Graduate Research Assistant

- Developed and enhanced **MERN**-based researcher and faculty portals, building **RESTful APIs** and interactive dashboards to modernize legacy research workflows, contributing to a **15%** increase in user engagement across research applications.
- Integrated researcher-facing applications with centralized **Identity and Access Management (IAM)** and **Single Sign-On (SSO)**, consolidating fragmented authentication workflows and contributing to a **15%** increase in faculty adoption.
- Researched and prototyped a **RAG**-based research assistant using **Python**, vector embeddings, prompt engineering, and LLM workflows to retrieve information from grant, compliance, and university research documents, reducing information retrieval time by **30%** compared with manual document lookup during prototype evaluation.
- Implemented application and infrastructure monitoring using **Prometheus** and **Grafana** across infrastructure supporting **80+ connected research devices**, analyzing telemetry to identify utilization patterns and resource optimization opportunities.

Hitachi Vantara (Bengaluru, India) 08/2022 – 08/2024
Software Engineer

- Developed scalable **RESTful microservices** for the **Lumada Data Catalog** using **Java** and **Spring Boot**, processing high-volume metadata across distributed data sources to improve catalog ingestion and query performance.
- Built resilient metadata ingestion and persistence workflows using **Spring Integration**, **JPA**, and **Hibernate**, applying **TDD**, **automated testing**, and **debugging** with fault-tolerant processing patterns to support **99.9%** service availability.
- Engineered asynchronous data migration and replication pipelines using **Spring Batch** and **Apache Kafka**, reducing synchronization bottlenecks when moving metadata across distributed cloud environments.
- Developed reactive data-access services using **Spring WebFlux** and integrated **React.js** components for metadata search and visualization workflows, reducing reporting data retrieval latency by **40%**.
- Automated staging deployment and multi-environment validation workflows using **Docker**, **Kubernetes**, **Jenkins**, **AWS**, and **Azure**, reducing regression-testing cycles by **60%** and improving release reliability across cloud environments.

Hitachi Vantara (Bengaluru, India) 01/2022 – 08/2022
Software Engineer (Co-Op)

- Developed **Java**-based data transformation plugins for the **Pentaho** platform, integrating relational **SQL** queries for data processing and optimizing query execution to improve reporting module efficiency by **15%**; prepared technical documentation and delivered **6+ demos** during Agile sprints.
- Built automated integration and interoperability test scripts using **Python** while assisting with containerized service deployments using **Docker** and internal **Kubernetes** staging clusters for **CI/CD** regression cycles.

Skills

Programming Languages:	Java, Python, C, C++, JavaScript, TypeScript, Go, SQL, HTML/CSS
Frameworks & Development:	Spring Boot, Spring WebFlux, Spring Integration, Spring Batch, JPA/Hibernate, React.js, Node.js, FastAPI, Django, REST APIs, Microservices
AI & Retrieval:	LangChain, LLMs, RAG, Prompt Engineering, Vector Search, Information Retrieval, Neo4j, Qdrant, BM25
Data & Messaging:	PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka
Cloud & DevOps:	AWS, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, Git, Maven, Gradle, CI/CD
Observability & Security:	Prometheus, Grafana, IAM, SSO, OAuth 2.0, RBAC
Core CS:	Data Structures, Algorithms, Object-Oriented Programming, Distributed Systems, Networking, TDD, Testing, Debugging, Agile, Scrum, SDLC

Projects & Certifications

- **ChronoMind Retrieval Engine:** Created a temporal memory reasoning engine that structures personal history as an evolving event graph, combining **Neo4j** for causal graph traversal and **Qdrant** for vector search; integrated **BM25 lexical retrieval** with a custom ranking pipeline to reconstruct chronological context timelines grounded in memory events.
- **Ride Matching Platform:** Engineered an enterprise-grade mobility ecosystem for real-time ride dispatching using a modular **full-stack architecture**; built live route visualizations, driver-matching workflows, and an **Admin Dashboard** for fleet monitoring, network analytics, and revenue tracking.
- **Claude Agentic AI Architect** certification (ongoing).